Lecture 11: Evolution, Emotion, and Reason: Emotions, Part I

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-11 (CC BY-NC-SA 3.0).

Before you start

This lecture begins a two-part look at the emotions from an evolutionary point of view. Professor Bloom argues that emotions are not the enemy of reason: they set our goals and priorities. He then looks at three cases: smiles, fear, and the feelings we have towards our relatives.

Learning goals

After this lecture you should be able to:

- Explain why the "Star Trek" view of emotions is wrong, using the cases of Phineas Gage and Elliot.
- Describe three types of smiles and give evidence that smiles are social signals.
- Explain why people fear snakes and spiders more than guns and cars.
- Explain how the "selfish gene" view can explain kindness towards kin (genetic relatives).
- Compare the "Cupboard Theory" and Bowlby's theory of why babies become attached to their mothers.

Key ideas

1. The Different Functions of Emotions

- An evolutionary view makes us ask questions about things that feel natural. William James wrote that only a psychologist would ask, "Why do we smile when pleased and not scowl?"
- Poop smells bad to us, but to a dung beetle it "smells just fine." Facts like this come from our biological nature, not from logic.
- Bloom starts with the "wrong theory" of emotions, shown in Star Trek: Mr. Spock and Data are very capable people without emotions.
- Steven Pinker, in How the Mind Works, argues that Spock must have had motives such as curiosity, fear and ambition. Without emotions, we would do nothing at all.

2. Phineas Gage and the Loss of Emotional Capacity

- In 1848, an iron rod went through the head of Phineas Gage, a construction foreman. He survived. He was not blind or deaf, and he did not lose language.
- But he "lost his character": he became rude and impatient, lost his family and his job, and ended up in the circus.

- People with damage to parts of the frontal cortex (the front of the brain) lose a large part of their emotions. They cannot prioritize.
- In Descartes' Error, Damasio describes a patient, Elliot, who lost much of his frontal lobe when a tumor was removed. He might spend a whole afternoon deciding how to sort files: by date or by size?
- Main point: emotions set goals and priorities. Without them, rationality fails. Emotions have universal roots (as Darwin knew), but cultures also differ.

3. Facial Expressions and Smiles in Particular

- Paul Ekman studies facial expressions; the class followed his instructions for an anger face. Faces show some differences across cultures, but also deep universals.
- Smiles are universal: children who are blind from birth smile, so smiling is not learned by watching faces. Nonhuman primates smile too.
- Smiles are social signals. Bowlers smile when they turn to their friends, not when they knock down the pins. Olympic gold medal winners smile when they face the crowd.
- The Duchenne smile shows real happiness, and the eyes move; only about one in ten people can fake it. The "Pan Am" smile is a greeting smile; the eyes do not move.
- Ten-month-olds give a real smile to their mother and a greeting smile to a stranger. John Gottman found that a true happiness smile between married couples is a better sign for the marriage. College yearbook smiles are related to how happy people are thirty years later.
- A third type, the coy or appeasement smile, shows embarrassment or stress: the person turns away and avoids eye contact. Rhesus monkeys and human babies show it.

4. Question and Answer on Smiles

Bloom explained "emotional contagion": it is very hard not to smile back at someone who smiles at you, and this is true for almost every emotion. He also guessed that details such as showing your teeth differ between countries.

5. Non-social Emotions: Fear

- Fear is basic and universal. It is "nonsocial": you can fear a person, but you can also fear falling off a cliff.
- Universal fears include spiders, snakes, heights, storms, darkness, blood, strangers and deep water. Few people fear guns, cars or electrical outlets.
- The reason: these things were dangerous to us through human evolution. Chimpanzees also develop phobias (very strong fears) of things like spiders and snakes.
- Children in inner-city Chicago said their two biggest fears were snakes and spiders.
- Judy DeLoache (University of Virginia) studies babies' fear of spiders and snakes. These are "pre-potent stimuli": things that naturally cause fear.

6. Social Emotions and Altruism

- Social emotions are felt towards kin or towards people we are not related to. "Altruism" is the biologists' word for kindness and generosity.

- Compare Gene "A" (care for your offspring) with Gene "B" (care only for yourself). Gene "A" wins. What matters is reproduction, not survival.
- From the germ's point of view, a cold virus uses your sneeze to spread itself. Richard Dawkins suggested that animals are vehicles that genes use to reproduce. So selfish genes can lead to altruistic animals.
- Haldane joked that he would give his life "for three brothers or five nephews or nine first cousins." How much we care follows genetic relatedness.
- Birds and mammals "invest in quality and not quantity." Because of how our brains are wired, a baby's cry annoys us and babies look cute (big forehead, upturned nose, chubby cheeks, big eyes). Adults with baby faces seem naive, kind and warm (the "baby-face bias").
- B.F. Skinner's "Cupboard Theory" says babies become attached to parents because parents give food. Bowlby says babies want comfort and social interaction. Harlow tested this with "wire mothers" (food) and "cloth mothers" (warmth, but no food). The results come in the next lecture.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
priority	something you decide is more important than other things	Elliot could not decide what to do first at work
frontal lobe	the front part of the brain	Gage and Elliot had damage here and lost much of their emotional capacity
Duchenne smile	a real smile of happiness that moves the eyes	Only about one in ten people can fake it
"Pan Am" smile	a polite greeting smile; the eyes do not move	Politicians often give this smile
coy (appeasement) smile	a smile of embarrassment or stress, with the head turned away	A baby smiles and turns its head away
emotional contagion	catching the emotion of the person in front of you	It is hard not to smile back at a smiling person
phobia	a very strong fear	Chimpanzees develop phobias of spiders and snakes
pre-potent stimuli	things that naturally cause fear	Spiders and snakes
altruism	kindness and generosity towards others	Parents caring for their children
kin	people who are genetically related to you	Haldane's brothers, nephews and cousins
baby-face bias	seeing adults with baby faces as naive, kind and warm	In mock trials, people with baby faces were more often found innocent

Check your understanding

1. What is the "wrong theory" of emotions shown in Star Trek, and why does Pinker think it makes no sense?
2. What did Phineas Gage lose after his accident, and what abilities did he keep?
3. What was Elliot's problem after his operation? What does his case show?
4. How can you tell a Duchenne smile from a "Pan Am" smile?
5. What do the studies of bowlers and Olympic winners show about smiling?
6. Why are most people more afraid of snakes and spiders than of guns and cars?
7. Explain Haldane's joke about "three brothers or five nephews or nine first cousins."
8. In Harlow's study, which "mother" should babies love if the Cupboard Theory is right? Which one if Bowlby is right?

Answer guide

1. It says we can function well without emotions. Pinker argues that Spock must have had motives such as curiosity and fear; without emotions he would do nothing.
2. He lost his character: he became rude and impatient and lost his job and family. He was not blind or deaf, and he did not lose language.
3. Elliot could not set goals or priorities; he could spend an afternoon deciding how to sort files. His case shows that without emotions, rationality fails.
4. In a Duchenne smile the eyes move, and it shows real happiness. In a Pan Am (greeting) smile, the eyes do not move.
5. Smiles are social signals. People smile when they want to communicate happiness to others, such as friends or a crowd.
6. Snakes and spiders were dangerous to us during human evolution, so we fear them naturally. Guns and cars were not dangers in our ancestral environment.
7. Saving several close relatives can save more copies of your genes than saving yourself. So genes that make us care for kin spread.
8. The Cupboard Theory says attachment comes from food, so babies should love the wire mother. Bowlby says babies want comfort, so they should love the cloth mother.

English practice

Useful phrases from the lecture

- "It turns out that..." - introduces a surprising finding. ("It turns out that this is a notion of how to think about the emotions that is deeply wrong.")
- "It's not as simple as that." - says an easy answer is too simple. Bloom used it about the idea that people smile because they are happy.
- "Avoid the temptation to say..." - warns against an easy but wrong answer.
- "From this perspective, ..." - introduces a new way of looking at something. ("From this perspective, a person is just a germ's way of making other germs.")

Speaking task (with your AI tutor)

- Explain why we need emotions to act rationally, in 4-6 sentences. Use: *priorities*, *frontal lobe*, *Phineas Gage*.
- Explain the three kinds of smiles, in 4-6 sentences. Use: *Duchenne smile*, *greeting smile*, *social signal*.

Writing task

- Write one paragraph (80-120 words): Why are people more afraid of snakes than of cars? Use the words "evolution" and "pre-potent stimuli", and give one study from the lecture.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 11. Ask me questions about the functions of emotions, one at a time."
- "Quiz me on the key vocabulary of Lecture 11."
- "I will explain the difference between a Duchenne smile and a Pan Am smile in my own words. Check my understanding and my English."
- "Ask me to explain Haldane's joke using the idea of the selfish gene."

Readings from the original course

- Gray, Peter. Psychology (5th edition), pp. 61-68, 345-351
- Pinker, Steven. "How the Mind Works." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 39-44

My notes